

PES1UG24AM126

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Brief Description

Allows a Faculty Researcher to record an equipment or consumable procurement expense against a sponsored grant. The system always performs a budget-balance check (an included use case) before the expense is committed, ensuring the grant is never overdrawn without explicit administrative override.

Epic

- **Epic ID / Name:** EPIC 1: Grant Budgeting & Procurement Expense Management
- **Description:** Enables faculty researchers to record procurement expenses against active sponsored grants with automated budget-balance validation, audit logging, and override workflows.

User Stories

Story 1: View Grant Budget & Spend Overview

- **Summary:** UC01.1: View Grant Budget & Spend Overview
- **User Story:** As a Faculty Researcher, I want to select an active grant and view its total allocation, category-wise spend, and remaining balance, so that I can assess available funds before making a procurement entry.
- **Priority:** High
- **Story Points:** 2

Story 2: Log Procurement Expense & Auto-Validate Balance

- **Summary:** UC01.2: Log Procurement Expense & Auto-Validate Balance
- **User Story:** As a Faculty Researcher, I want to submit expense details (amount, category, vendor, invoice reference) and have the system automatically run a budget-balance check, so that approved entries commit to the budget, update the remaining balance, and record an immutable audit-ledger entry.
- **Priority:** Highest
- **Story Points:** 5

Story 3: Request Dean Override for Exceeded Budget

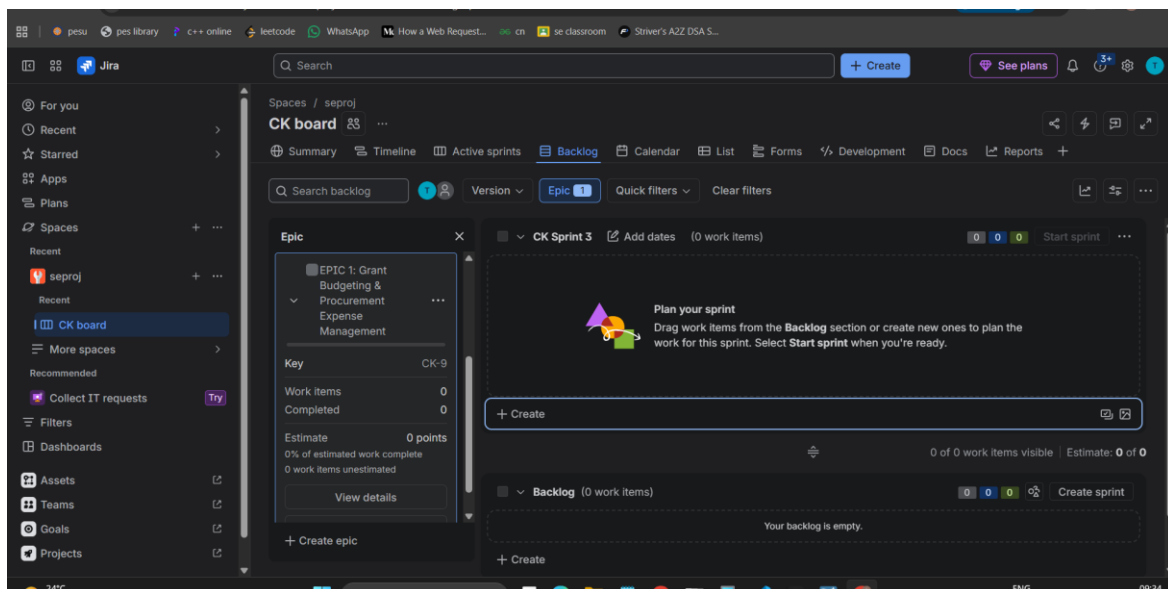
- **Summary:** UC01.3: Request Dean Override for Exceeded Budget
- **User Story:** As a Faculty Researcher, I want to be notified of budget/category shortfalls and given the option to route an override request to the Research Dean, so that critical procurement isn't entirely blocked when funds are exceeded.

- **Priority:** Medium
- **Story Points:** 3

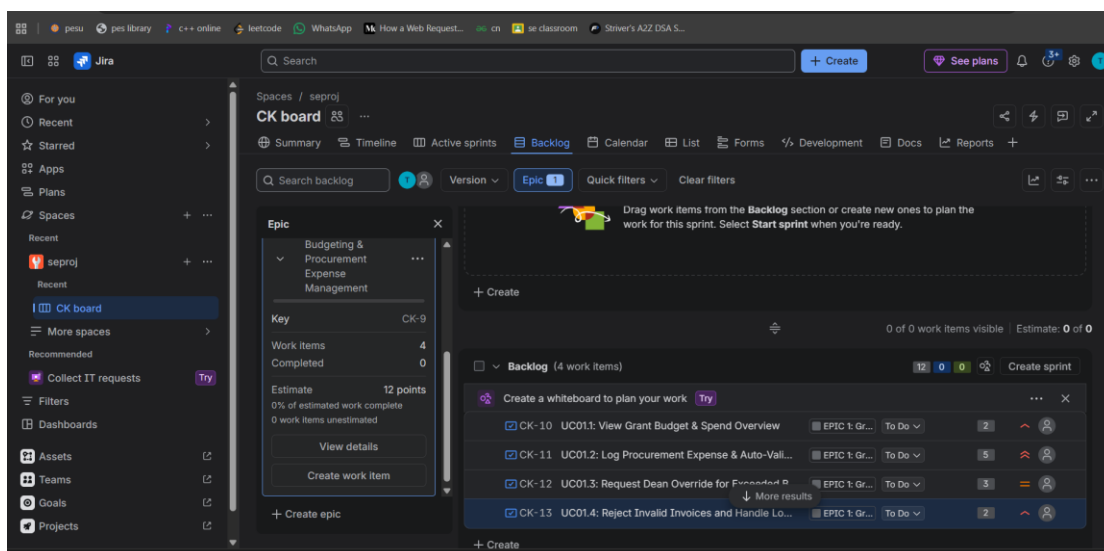
Story 4: Reject Invalid Invoices and Handle Locked Grants

- **Summary:** UC01.4: Reject Invalid Invoices and Handle Locked Grants
- **User Story:** As a System, I want to reject entries with missing/malformed invoice references and abort transactions on locked grants, so that data integrity and compliance are maintained.
- **Priority:** High
- **Story Points:** 2

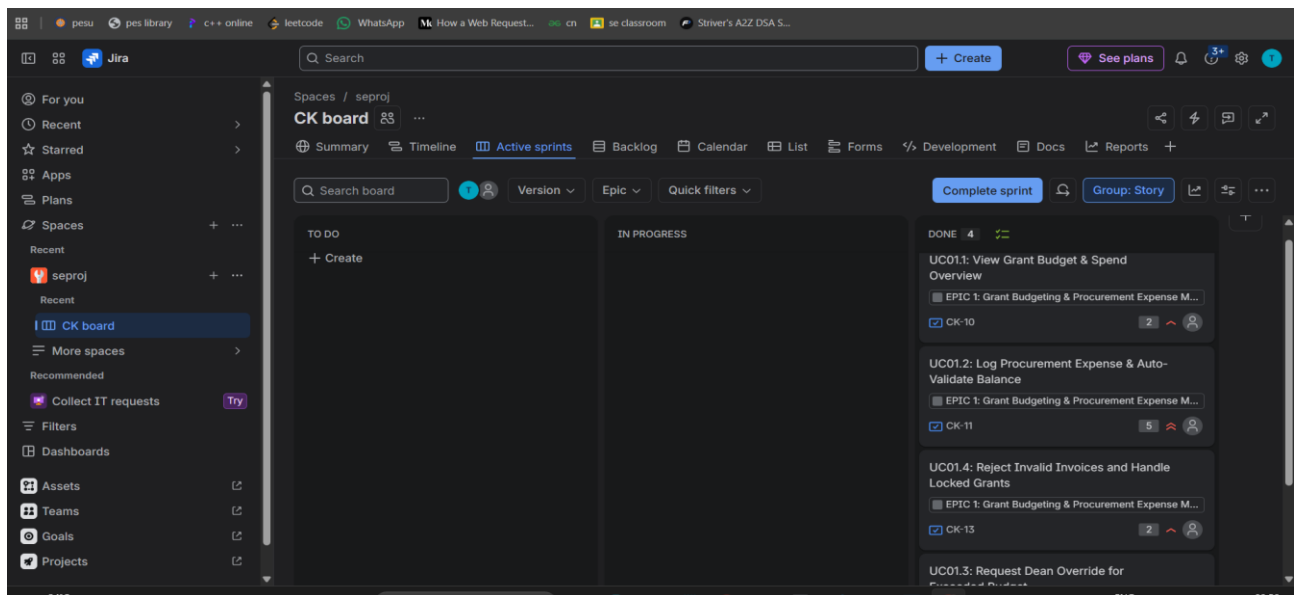
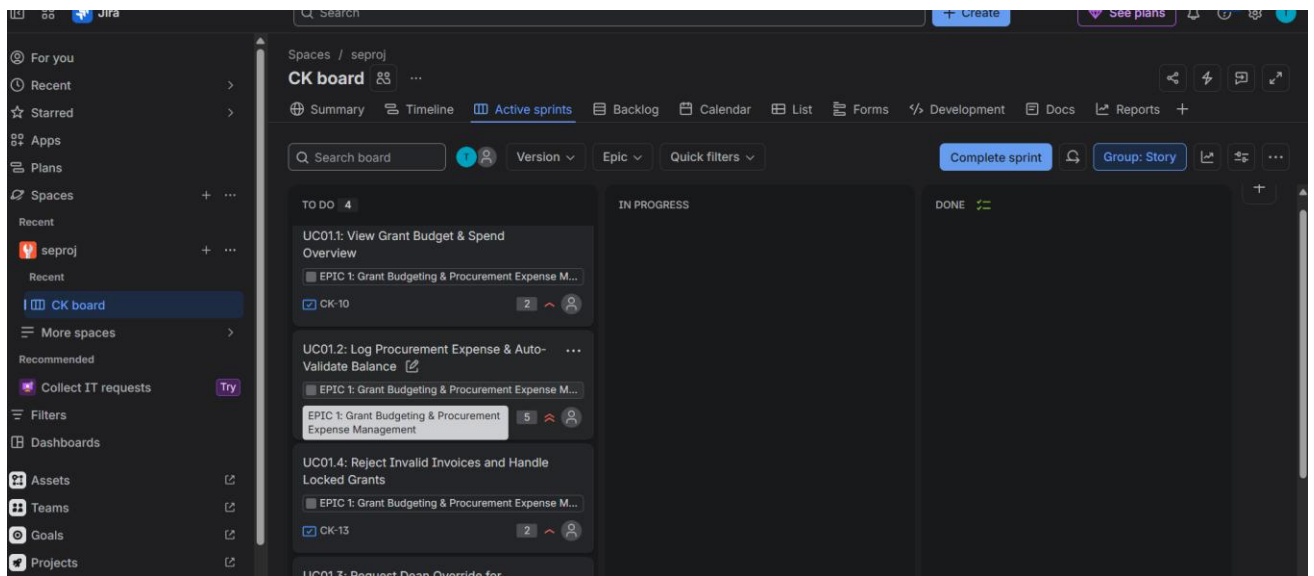
Project Setup & Epic Creation



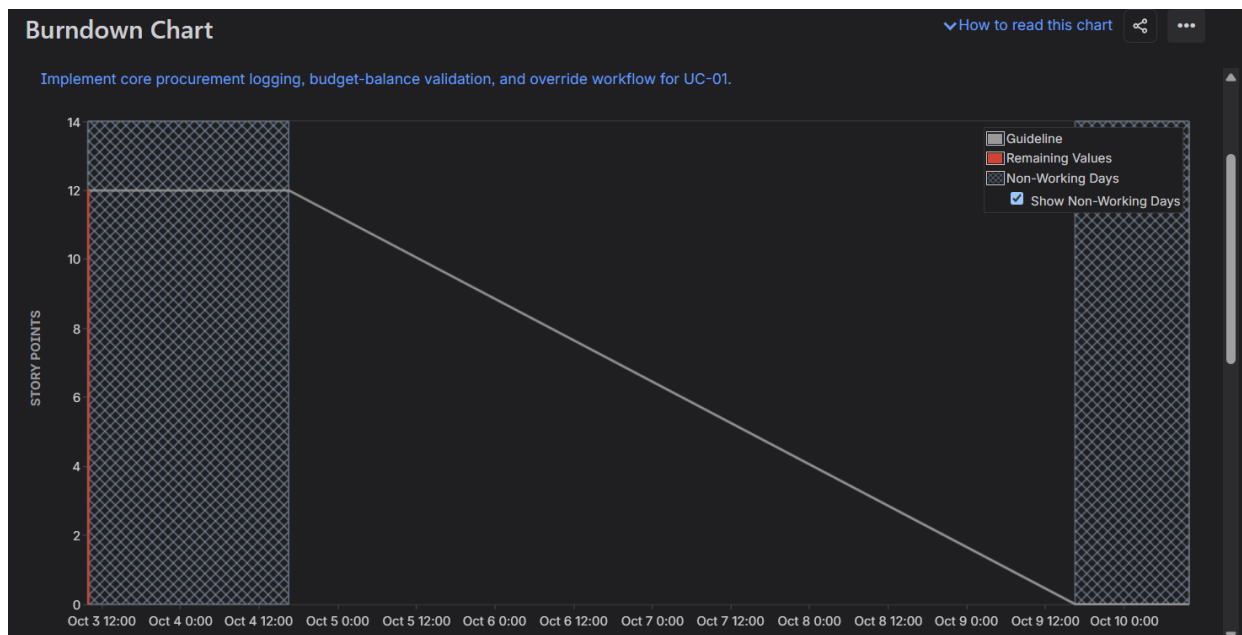
User Story Creation, Prioritization, and Estimation



Sprint Planning & Execution



Reporting & Analytics



- **Did your estimations reflect the actual effort?**

Yes. Assigning higher story points (5) to the automated budget check accurately reflected the risk and complexity of integrating the included `Check Budget Balance` use case and audit-ledger write.

- **Was your backlog well-prioritized?**

Yes. Core features like spend visibility and balance validation were set to `Highest` and `High` priority, while edge-case workflows like administrative overrides were prioritized as `Medium`.

- **How did your simulated sprint align with your plan?**

The sprint execution successfully completed all planned story points (12 points total) without scope creep, ensuring all acceptance criteria for UC-01 were met within the time-box.

- **What insights did the burndown chart give about your team's capacity?**

The burndown chart demonstrated steady progress from 12 remaining points down to 0, confirming that a 12-point velocity is achievable for the team in a single mini-sprint cycle.